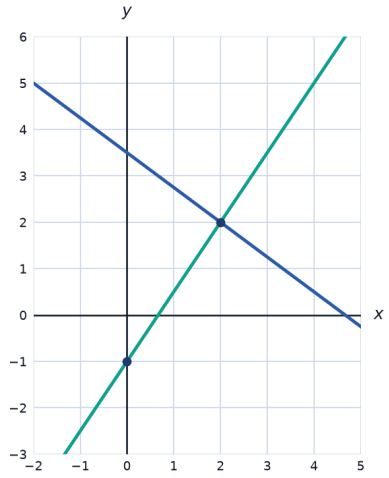
Elementary & Intermediate Algebra

Section 1: Linear Equations & Fractional Coefficients

1 Solve the linear equation for x: $\frac{2x-3}{4} - \frac{x+1}{3} = \frac{5}{12}$.



a. 7

b. 8

c. 9

d. 11

2 If $5(3x - 2) + 4 = 2(6x + 7) - 3$, what is the value of x?

a. 5

b. 6

c. 7

d. 8

3 For what value of k does the equation $4(2x - 5) + k = 8x - 13$ have infinitely many solutions?

a. 7

b. 13

c. 20

d. -7

Section 2: Linear Systems & Inequalities

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4 If $3x - 2y = 19$ and $2x + 5y = 0$, what is the value of $x + y$?

a. 3

b. 5

c. 7

d. 9

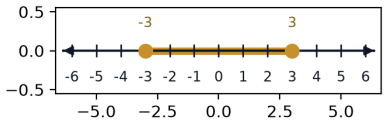
5 What is the complete solution set for the compound inequality $-7 \leq 3x + 2 < 11$?

a. $-3 \leq x < 3$

b. $-5 \leq x < 4$

c. $-3 < x \leq 3$

d. $x \geq -3$



6 A company sells widgets for 14 each with fixed costs of $1,800$ and variable costs of 5 per widget. How many widgets must be sold to make a profit of at least 900?

a. 200

b. 250

c. 300

d. 350

Section 3: Factoring Quadratics & Quadratic Formula

7 What are the solutions to the quadratic equation $2x^2 - 7x - 15 = 0$?

a. $x = 5$ and $x = -\frac{3}{2}$

b. $x = -5$ and $x = \frac{3}{2}$

c. $x = 3$ and $x = -\frac{5}{2}$

d. $x = 5$ and $x = 3$

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8 What is the sum of the solutions of the equation $3x^2 - 12x + 5 = 0$?

a. -4

b. 4

c. $\frac{5}{3}$

d. 12

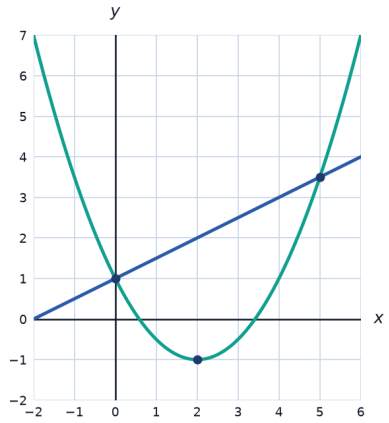
9 For what value of c will the quadratic equation $x^2 - 10x + c = 0$ have exactly one distinct real solution?

a. 5

b. 10

c. 25

d. 100



Section 4: Rational Expressions & Radical Equations

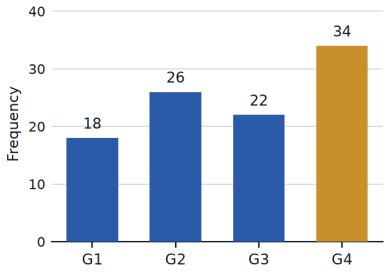
10 Simplify the rational expression: $\frac{x^2 - 9}{2x^2 + 5x - 3}$ for all $x \neq \frac{1}{2}, -3$.

a. $\frac{x - 3}{2x - 1}$

b. $\frac{x + 3}{2x + 1}$

c. $\frac{x - 3}{2x + 1}$

d. $\frac{1}{2x - 1}$



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11 Solve the radical equation for all valid real values: $\sqrt{3x + 13} = x + 3$.

a. $x = 1$ only

b. $x = -4$ only

c. $x = 1$ and $x = -4$

d. No real solutions

12 If $\frac{1}{a} + \frac{1}{b} = \frac{5}{12}$ and $ab = 24$, what is the value of $a + b$?

a. 5

b. 10

c. 12

d. 15

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